



IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

# IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

## Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume

**CAPTURE SOURCE / INGESTION INTERFACE**  
Forensic Uploaded PCAP: sih26\_asim\_golden.pcap

**ANALYSIS TIMEFRAME WINDOW**  
Forensic Session Duration: 0.00s – 142.08s (...)

### INITIATOR GATEWAY

#### Company Branch Gateway 1

OS: Ubuntu Linux 24.04  
Interface: tailscale0 / eth0  
Private IP: 100.119.32.83  
VPN Daemon: StrongSwan v5.9.8

IPsec ESP

ESP over UDP 4500 (NAT-T) · Tunnel Mode...

### RESPONDER GATEWAY

#### Enterprise Datacenter Core

OS: Ubuntu Linux 24.04  
Interface: tailscale0 / eth0  
Private IP: 100.127.207.119  
VPN Daemon: StrongSwan v5.9.8

|                                                       |                                                           |                                                          |                                                          |                                                          |                                                                 |
|-------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------|
| <b>TOTAL PACKETS</b><br><b>557</b><br>Frames Ingested | <b>FLOW VOLUME</b><br><b>22.8 MB</b><br>Encrypted Payload | <b>AVG RATE</b><br><b>4 pkt/s</b><br>Throughput Velocity | <b>PEAK RATE</b><br><b>1,120 pkt/s</b><br>Observed Burst | <b>INTERFACE MTU</b><br><b>1500 bytes</b><br>UP / ACTIVE | <b>DROPS / ERRORS</b><br><b>0 (0.00%)</b><br>Monotonic Verified |
|-------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------|

## Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs

### IKE PROTOCOL EXCHANGES & HANDSHAKE SEQUENCE

IKE\_SA\_INIT [COMPLETED / SUCCESS] Initiator/Responder DH public keys and nonces exchanged. Enforced ECP..  
IKE\_AUTH [COMPLETED / SUCCESS] Mutual authentication executed via Pre-Shared Key. Initial Child SA e..  
CREATE\_CHILD\_SA [MONITORED] Child SA rekeying evaluated. No ephemeral DH transform negotiated (PF..

#### ACTIVE SECURITY ASSOCIATION (SPI) PAIRS

|                                            |                                            |
|--------------------------------------------|--------------------------------------------|
| <b>INITIATOR SPI</b><br>0x8b14e9f28a1c9034 | <b>RESPONDER SPI</b><br>0x4a7c10b83f09de21 |
| <b>INBOUND ESP SPI</b><br>0xc0a80102       | <b>OUTBOUND ESP SPI</b><br>0xc0a80103      |

#### NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

| TRANSFORM TYPE                | IKE SA NEGOTIATED      | CHILD SA NEGOTIATED    | COMPLIANCE STATUS | BENCHMARK STANDARD   |
|-------------------------------|------------------------|------------------------|-------------------|----------------------|
| Encryption Algorithm          | AES-256-GCM            | AES-256-GCM            | APPROVED ...      | RFC 8221 / CNSA 2.0  |
| Key Exchange (Diffie-Hellman) | ECP-256                | ECP-256                | APPROVED ...      | NIST SP 800-77 §4.2  |
| Pseudo-Random Function (PRF)  | HMAC-SHA2-256          | N/A (AEAD Suite)       | APPROVED ...      | RFC 7296             |
| Integrity / Authentication    | Built-in (128-bit ICV) | Built-in (128-bit ICV) | APPROVED ...      | NIST SP 800-77 Rev 1 |



## DATA-PLANE TELEMETRY & AI/ML PAYLOAD CLASSIFICATION

# Data-Plane Stream Telemetry & Inference

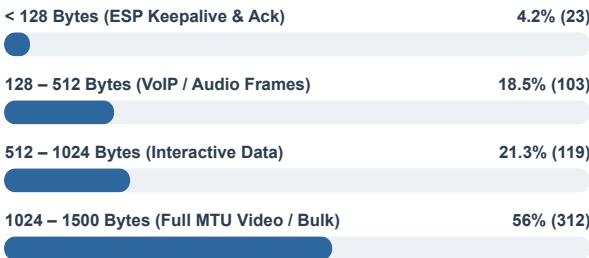
Packet size distribution histograms, timing analysis, sequence progression, and inner encrypted traffic classification

## Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression

### PACKET SIZE DISTRIBUTION HISTOGRAM

BINNED MTU



### INTER-ARRIVAL TIME & SEQUENCE INTEGRITY

RFC 4303

MEAN IAT  
2.36 ms

MEDIAN IAT  
1.82 ms

JITTER ( $\sigma$ )  
0.48 ms

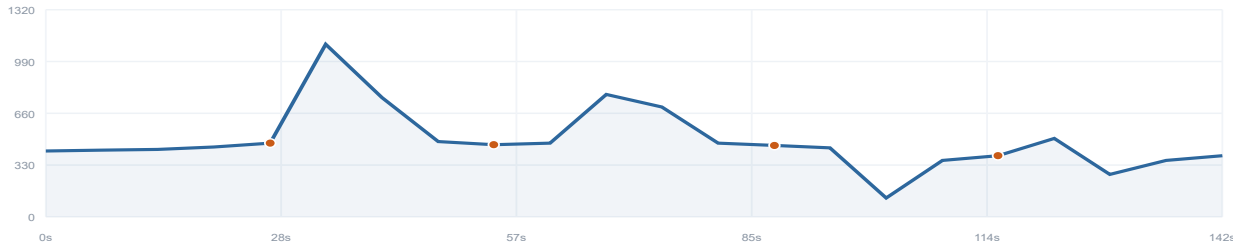
MIN / MAX  
0.12 ms / 18.40 ms

Sequence Monotonicity: 100.0% Strict Monotonic | Out of Order: 0 packets (0.00%)

Rollover Protection: SAFE (64-bit Extended Sequence Numbers active)

Anti-Replay Verification: RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted

### Throughput & Packet Velocity Timeline



#### Traffic Spike (00:15 – 00:18)

Burst of high-throughput encrypted packet activity.

#### Large Transfer (00:27 – 00:30)

Elevated data payload transmission cluster.

#### Throughput Dip (00:41 – 00:...

Activity briefly stabilized to baseline keepalive rate.

#### Session Steady (00:49 – 00:56)

Continuous isochronous frame cadence observed.

## Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals

### OPERATING MODE DETERMINATION

99.4% CONFIDENCE

#### IPsec Tunnel Mode

Encapsulating Security Payload (ESP) header envelopes internal IPv4 header with distinct external overlay IP routing.

SHANNON ENTROPY  
7.98 / 8.0

CLUSTERING COEFF  
0.84

BURSTINESS RATIO  
1.42

### INNER ENCRYPTED TRAFFIC CLASSIFICATION

94.2% CONFIDENCE

Encrypted Video Stream 94.2% [CI: 92.1% – 96.3%]

VoIP / Realtime Audio 3.1% [CI: 1.9% – 4.3%]

Bulk Data & Sync 1.8% [CI: 0.8% – 2.8%]

Web / Interactive 0.9% [CI: 0.2% – 1.6%]



ENGINEERING REMEDIATION & RAW FORENSIC EVIDENCE

# Implementation Patches & Trace Index

Actionable gateway configuration blocks ready for deployment, and raw packet trace appendix with flow JSON dumps

## Section 5: Implementation Remediation Patches

Actionable configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)

```
# 1. StrongSwan Gateway Patch
(/etc/swanctl/conf.d/apex_hardened.conf)

# /etc/swanctl/conf.d/apex_hardened.conf
connections {
    apex-ipsec-hardened {
        version = 2
        local_addrs = 100.119.32.83
        remote_addrs = 100.127.207.119
        proposals = aes256gcm16-ecp256-sha256

        children {
            apex-child {
                # STRICT MANDATE: Enforce DH Group 19 (ECP
                -256) on Child SA for Perfect Forward Secrecy
                esp_proposals = aes256gcm16-ecp256!
                replay_window = 64
                life_time = 3600s
                life_bytes = 1000000000
                dpd_action = restart
            }
        }
    }
}
```

```
! 2. Cisco IOS-XE / ASA Enterprise CLI Configuration Block

! Cisco IOS-XE / ASA Enterprise Gateway Remediation
crypto ikev2 proposal APEX_IKE_PROP
  encryption aes-gcm-256
  prf sha256
  group 19
!
crypto ipsec transform-set APEX_ESP_SET esp-gcm 256
  mode tunnel
!
crypto ipsec profile APEX_IPSEC_PROFILE
  set transform-set APEX_ESP_SET
  set pfs group19
  set security-association replay window-size 64
  set security-association lifetime seconds 3600
```

**VALIDATION COMMANDS:** `$ swanctl --load-all && swanctl --initiate --child apex-child`    `$ swanctl --list-sas | grep -E 'AES_GCM|ECP_2'`

## Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

| PKT # | OFFSET  | LAYER / PROTO | TYPE / EXCHANGE    | SPI IDENTIFIER          | LENGTH | DISSECTION SUMMARY                      |
|-------|---------|---------------|--------------------|-------------------------|--------|-----------------------------------------|
| #1    | +0.000s | UDP 500       | IKE_SA_INIT (Req)  | Initiator=0x8b14e9f2... | 384 B  | DH Group 19 proposal exchange           |
| #2    | +0.014s | UDP 500       | IKE_SA_INIT (Resp) | Responder=0x4a7c...     | 384 B  | DH public key confirmation              |
| #3    | +0.028s | UDP 4500      | IKE_AUTH (Req)     | Initiator=0x8b14e9f2... | 448 B  | PSK mutual identity authentication      |
| #4    | +0.042s | UDP 4500      | IKE_AUTH (Resp)    | Responder=0x4a7c...     | 448 B  | Child SA creation and Traffic Selectors |

RAW FLOW TELEMETRY JSON DUMP

RFC 8221 & SIEM COMPATIBLE

```
{
  "session_id": "AV-20260916-TECH-058D",
  "protocol": "IPsec ESP / IKEv2",
  "endpoints": {
    "initiator": "100.119.32.83",
    "responder": "100.127.207.119"
  },
  "ike_transforms": {
    "encryption": "AES-256-GCM",
    "dh_group": "ECP-256",
    "prf": "HMAC-SHA256"
```